

# Louis Euyseok Han

[www.louis-han.info](http://www.louis-han.info) | [louisshan@seas.upenn.edu](mailto:louisshan@seas.upenn.edu) | (267) 456-5667 | Canadian Citizen

## Education

|                                                                                                                                                                                                                  |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>University of Pennsylvania</b> , School of Engineering and Applied Science, Philadelphia, PA<br>M.S.E., Mechanical Engineering and Applied Mechanics ( <b>Concentration: Mechatronic and Robotic System</b> ) | May 2027 |
| <b>Seoul National University</b> , Seoul, Korea<br>B.S., Mechanical and Aerospace Engineering                                                                                                                    | Feb 2017 |

## Skills

**Programming & Data Engineering:** Python, C++, SQL, Shell Scripting, Linux, Test-Driven Development, Docker  
**Project Management & Collaboration:** Git, GitHub Actions (CI/CD), Atlassian Jira, Confluence, Notion  
**CAD Tools & Prototyping:** SolidWorks, 3D Print(Formlabs), Laser Cutting, Shop Tools, Rapid Prototyping  
**Mechanical & Electrical:** Mechatronics, Feedback Control, Manufacturing, Materials, ROS2, System Integration  
**Soft Skills:** MS Word, Excel, PowerPoint, Problem Solving, Team Leadership, Public Speaking, Patent Prosecution

## Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>University of Pennsylvania</b> , Philadelphia, PA<br>Research Assistant @ Micro Sensors & Micro Actuators Group (Prof Mark G. Allen)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Oct 2025 - Present                       |
| <ul style="list-style-type: none"><li>Developed and experimentally evaluated an aluminum-air battery system for sustainable robotics</li><li>Designed and prototyped flow-cell system for controlled electrolyte circulation to improve Al-air battery performance</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| <b>Korea Advanced Institute of Science and Technology (KAIST)</b> , Daejeon, Korea<br>Research Intern @ Advanced Machine Intelligence Lab (Prof Tae-hyun Oh)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Jun 2025 - Aug 2025                      |
| <ul style="list-style-type: none"><li>Built a pipeline with PyTorch, NumPy, and vision-language models to predict material properties of objects from RGB and NIR images. Integrated the estimated values into 3D point clouds to create property-annotated 3D models</li></ul>                                                                                                                                                                                                                                                                                                                                                                                             |                                          |
| <b>Patsol</b> , Seoul, Korea<br>Software Developer, Cofounder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Oct 2024 - Jan 2025                      |
| <ul style="list-style-type: none"><li>Architected and implemented end-to-end RAG patent-search pipeline using Python, HuggingFace, Elasticsearch, enabling inventors to query patents via natural language instead of inefficient Boolean keyword filters</li><li>Led roadmap presentation to SpringCamp VC and secured \$71K seed funding by demonstrating scalable retrieval architecture and subscription-based monetization strategy</li></ul>                                                                                                                                                                                                                          |                                          |
| <b>Yogiyo (Food Delivery Platform, formerly Delivery Hero Korea)</b> , Seoul, Korea<br>Back-end Software Developer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Dec 2022 - Sep 2024                      |
| <ul style="list-style-type: none"><li>Improved portal websites and self-service tools for 500,000+ restaurant owners and optimized internal backoffice tools for Yogiyo staff using RESTful APIs, automated testing, and CI/CD, reducing ~40K monthly manual support tickets</li><li>Automated ingestion of ~1M daily food-delivery orders with Airflow, eliminating manual workflows; collaborated with SRE to scale compute resources for ETL workloads</li><li>Designed pre-aggregated order datasets and optimized database schemas to reduce query latency (from 20s → ~2s). Built scalable Django REST analytics APIs to power order-based sales dashboards</li></ul> |                                          |
| <b>Korean Intellectual Property Office (KIPO)</b> , Daejeon, Korea<br>Patent Examiner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jun 2022 - Dec 2022, Apr 2018 - Feb 2019 |
| <ul style="list-style-type: none"><li>Conducted prior-art searches and substantive examination for 40+ patent applications per month, evaluating novelty, inventive step, and industrial applicability under Korean patent law</li><li>Drafted numerous Office Actions, including Notices of Reasons for Rejection and Final Rejections</li></ul>                                                                                                                                                                                                                                                                                                                           |                                          |

## Technical Projects

|                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wharton AI Hackathon (Hack-AI-thon)</b> (Python, Machine Learning, Prompt Engineering)                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Advanced to the Hackathon finals by developing an AI-powered hotel review question generation system for Expedia using machine learning, generating targeted follow-up questions to elicit more specific and diverse feedback.</li></ul> |
| <b>Dual Motor Autonomous Car with Web Dashboard</b> (ESP32, C++, PID Control, JS, SolidWorks, ToF Sensors)                                                                                                                                                                     |
| <ul style="list-style-type: none"><li>Built an ESP32-based dual-motor robotic car featuring PID speed control, a real-time Wi-Fi web dashboard, and autonomous point-to-point navigation using ToF-based wall detection on a custom SolidWorks-designed chassis</li></ul>      |

## Leadership Experience

|                                                                                                                                                    |                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>UPenn Garage Lab</b> (machine shop and prototyping facility), Lab Instructor                                                                    | Oct 2025 - Present  |
| <b>UPenn Mechanical Engineering Graduate Association</b> , Vice President                                                                          | Sep 2025 - Sep 2026 |
| <b>APEC, Intellectual Property Expert Sub-Group Delegate</b>                                                                                       | Aug 2018            |
| <ul style="list-style-type: none"><li>Represented the Republic of Korea in APEC intellectual property meetings as a KIPO Patent Examiner</li></ul> |                     |